

COURSE NOTES

Abstract Algebra III

Instructor:

Prof. Zhicheng FENG

Student:

Leo

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Chapter 1

Lecture 1: Modules

Message: QQ, BB. HW 30%, Mid 20%, Final 50%

1.1 Basic Definitions and Examples

Definition 1.1.1. Let R be a ring with 1, and let M be an abelian group. We say that M is a left R -module if R “left acts” on M :

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm$$

such that

$$\begin{aligned} r(m + n) &= rm + rn, \\ (r + s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m. \end{aligned} \quad \text{for any } r, s \in R, m, n \in M$$

A **right R -module** is defined by $(r, m) \mapsto mr$ with the axiom $m(rs) = (mr)s$. If R is commutative, left-modules and right-modules coincide. Otherwise, one can transform using the opposite ring R^{op} .

Consider M an R -module, then $\forall r \in R$, $\varphi_r : M \rightarrow M$ by $\varphi_r(m) = rm$ is an abelian group homomorphism. Let

$$\text{End}(M) = \{\varphi : M \rightarrow M \text{ is a abelian group homomorphism} \}$$

Then $\text{End}(M)$ is a ring with composition as the multiplication.

Theorem 1.1.2. Let M be an abelian group. Then M is an R -module if and only if there is a ring homomorphism $R \rightarrow \text{End}(M)$.

Proof. ($\Rightarrow$) Suppose M is an R -module. Define $\rho : R \rightarrow \text{End}(M)$ by

$$\rho(r)(m) = rm, \quad \forall r \in R, m \in M.$$

For each $r \in R$, the map $\rho(r) : M \rightarrow M$ is a group homomorphism since $r(m_1 + m_2) = rm_1 + rm_2$. Moreover, ρ is a ring homomorphism:

- $\rho(r + s)(m) = (r + s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r + s) = \rho(r) + \rho(s)$.
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$.
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \text{id}_M$.

($\Leftarrow$) Suppose $\rho : R \rightarrow \text{End}(M)$ is a ring homomorphism. Define the scalar multiplication $R \times M \rightarrow M$ by

$$(r, m) \mapsto \rho(r)(m).$$

We verify the module axioms:

- $r(m_1 + m_2) = \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2) = rm_1 + rm_2$.
- $(r + s)m = \rho(r + s)(m) = (\rho(r) + \rho(s))(m) = \rho(r)(m) + \rho(s)(m) = rm + sm$.
- $(rs)m = \rho(rs)(m) = (\rho(r) \circ \rho(s))(m) = \rho(r)(\rho(s)(m)) = r(sm)$.
- $1m = \rho(1)(m) = \text{id}_M(m) = m$.

Hence M is an R -module. □

Example 1.1.3. (1) A vector space is a module over a field.

(2) An abelian group is a module over the ring of integers $\mathbb{Z}$.

$$\mathbb{Z} \times M \rightarrow M, \quad (n, m) \mapsto n \cdot m := m + m + \cdots + m \quad (n \text{ times})$$

(3) An ideal I of a ring R is a module over R .

(4) Let M be an abelian group. Then M is an $\text{End}(M)$ -module via

$$\text{End}(M) \times M \rightarrow M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K , and let $A : V \rightarrow V$ be a linear transformation. Then V is a $K[x]$ -module via

$$K[x] \times V \rightarrow V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.

(7) Let R be a ring with 1 and $A \in M_n(R)$. The solution of $AX = 0$ where $X \in R^n$ is a right R -module.

Definition 1.1.4 (R -submodule). Let R be a ring (with 1) and let M be a left R -module. A subset $N \subseteq M$ is called an R -submodule of M if:

1. $0 \in N$;
2. $n_1, n_2 \in N \implies n_1 + n_2 \in N$;

3. $n \in N \implies -n \in N$;
4. $r \in R, n \in N \implies rn \in N$.

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R -action.

Example 1.1.5. (1) Let V be a vector space over a field K , then any subspace of V is a k -submodule of V .

(2) Let G be an abelian group, then any subgroup of G is a $\mathbb{Z}$ -submodule of G .

(3) Let R be a ring, ${}_R R$ is a left (regular) R -module. Then any left ideal of R is a R -submodule of ${}_R R$.

(4) Let G be an abelian group, which is an $\text{End}(G)$ -module. Then any full invariant subgroup of G is a $\text{End}(G)$ -submodule of G .

Question: Are submodules of a finitely generated R -module finitely generated?

Answer: No, not in general. This property holds if and only if the ring R is Noetherian.

Example 1.1.6 (Submodule of a f.g. module that is not f.g.). Let k be a field and consider the polynomial ring $R = k[x_1, x_2, x_3, \dots]$ in infinitely many variables.

- The ring R itself can be viewed as a left R -module, which we denote by $M = R$. M is finitely generated, for instance by the single element $1 \in R$.
- Consider the ideal $I = \langle x_1, x_2, x_3, \dots \rangle$ generated by all the variables. This ideal I is a submodule of $M = R$.
- This submodule I is not finitely generated. To see this, suppose for contradiction that I is finitely generated by a set of polynomials $\{f_1, \dots, f_n\}$. Let N be the largest index of any variable appearing in any of the polynomials $f_1, \dots, f_n$. Then any element in the ideal generated by $\{f_1, \dots, f_n\}$ must be a polynomial in variables $x_1, \dots, x_N$. However, the polynomial x_{N+1} is in I but cannot be expressed as a combination of $f_1, \dots, f_n$. This is a contradiction.

Therefore, the ideal I is a submodule of the finitely generated R -module R which is not itself finitely generated. This shows that the ring R is not Noetherian.

Definition 1.1.7. Let M be an R -module, $M_i, i \in I$ are R -submodules of M . Then

$$\bigcap_{i \in I} M_i$$

is an R -submodule of M .

$$\sum_{i \in I} = \left\{ \sum_{i \in I \text{ finite}} m_i \mid m_i \in M_i \right\}$$

is an R -submodule of M .

Definition 1.1.8. Let M be an R -module, $S \subseteq M$ a subset.

$$\langle S \rangle = \left\{ \sum_{\text{finite}} a_i s_i \mid a_i \in R, s_i \in S \right\}$$

or

$$\langle S \rangle = \bigcap_{\substack{N \text{ is an } R\text{-submodule of } M, \\ S \subseteq N}} N$$

is an R -submodule of M , called the R -submodule of M generated by S .

If $M = \langle S \rangle$, then we say M is generated by S . In particular, if $|S| = 1$ then we call M a cyclic module. If $|S| < \infty$ then we call M a finite generated module.

Definition 1.1.9. Let R be a ring and let M be a left R -module. We say that M is a Noetherian module if every ascending chain of submodules

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots.$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. Let R be a ring, let M be a left R -module, and let N be a submodule of M . The quotient module (or factor module) of M by N is the quotient abelian group

$$M/N = \{x + N \mid x \in M\},$$

with scalar multiplication defined by

$$r(x + N) = rx + N \quad (r \in R, x \in M).$$

This is well-defined, and with this operation M/N becomes a left R -module.

Example 1.2.2. (1) Let R be a ring and an R -module, I be an ideal of R , then R/I is a quotient R -module of R by I .

(2) Consider the ring $\mathbb{Z}$ and the module $\mathbb{Z}$ over $\mathbb{Z}$, then for $n \in \mathbb{Z}$, $n\mathbb{Z}$ is a submodule of $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$ -module of $\mathbb{Z}$ by $n\mathbb{Z}$. Furthermore, $\mathbb{Z}/n\mathbb{Z}$ is a cyclic module.

Definition 1.2.3. Let R be a ring, and let M and N be left R -modules. A group homomorphism

$$f : M \rightarrow N$$

is called an R -module homomorphism (or R -linear map) if for all $x, y \in M$ and all $r \in R$, one has

$$f(rx) = rf(x) \quad \forall r \in R, x \in M.$$

Definition 1.2.4. Let f be an R -module homomorphism. The kernel of f is the set

$$\ker(f) = \{x \in M \mid f(x) = 0\}.$$

$\ker(f)$ is a submodule of M . The image of f is the set

$$\operatorname{im}(f) = \{f(x) \in N \mid x \in M\}.$$

$\operatorname{im}(f)$ is a submodule of N .

If $\ker(f) = 0$, then f is injective. If $\operatorname{im}(f) = N$, then f is surjective. If $\ker(f) = 0$ and $\operatorname{im}(f) = N$, then f is bijective.

A bijective R -module homomorphism is called an isomorphism.

Example 1.2.5. Let R be a ring and M, N be R -modules. Then $\operatorname{Hom}_R(M, N)$ is an R -module when R is commutative.

Theorem 1.2.6. If $f : M \rightarrow N$ is an R -module homomorphism, then we have a canonical R -module isomorphism:

$$M/\ker(f) \rightarrow \operatorname{im}(f)$$

Theorem 1.2.7. Let N be an R -submodule of M , $\pi_N : M \rightarrow M/N$ be a quotient map. Then π_N induces a natural bijection:

$$\{R\text{-submodules of } M \text{ containing } N\} \rightarrow \{R\text{-submodules of } M/N\}$$

Theorem 1.2.8. If $N \subseteq H$ are two R -submodules of M , then

$$M/H \cong M/N/(H/N),$$

Theorem 1.2.9. If N, H are R -submodules of M , then

$$\frac{H+N}{N} \cong \frac{H}{H \cap N}$$

Proposition 1.2.10. *Let M be an R -module, N a submodule of M , M' an R -module, and $\varphi : M \rightarrow M'$ a morphism such that $N \subseteq \ker(\varphi)$. Then there exists a unique morphism*

$$\varphi_N : M/N \rightarrow M'$$

such that the following diagram commutes. Moreover, $\ker(\varphi_N) = \ker \varphi / N$.

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_N \downarrow & \nearrow \varphi_N & \\ M/N & & \end{array}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R -module,*
- *$\theta : M \rightarrow S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \rightarrow S$ such that the following diagram commutes.

$$\begin{array}{ccc} M & \xrightarrow{\theta} & S \\ \pi_N \downarrow & \nearrow \exists! \psi & \\ M/N & & \end{array}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \rightarrow M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \rightarrow M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N , and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \rightarrow S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute

$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$

But the identity morphism $\text{id}_{M/N} : M/N \rightarrow M/N$ also satisfies

$$\text{id}_{M/N} \circ \pi_N = \pi_N.$$

By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that

$$u \circ \psi = \text{id}_{M/N}.$$

Similarly,

$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$

But the identity morphism $\text{id}_S : S \rightarrow S$ also satisfies

$$\text{id}_S \circ \theta = \theta.$$

By the uniqueness part of the universal property of (S, θ) , we obtain

$$\psi \circ u = \text{id}_S.$$

Therefore ψ is an isomorphism, with inverse u .

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism

$$\psi : M/N \rightarrow S$$

such that

$$\psi \circ \pi_N = \theta.$$

Hence there exists a unique isomorphism

$$\psi : M/N \rightarrow S$$

for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism $\bar{\varphi} : \text{coim } \varphi \rightarrow \text{im } \varphi$ such that the following diagram commutes:*

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_{\ker(\varphi)} \downarrow & & \uparrow \\ \text{coim}(\varphi) & \xrightarrow{\bar{\varphi}} & \text{im}(\varphi) \end{array}$$

Recall:

$$\begin{aligned}\varphi : M &\rightarrow M', \quad R\text{-morphism} \\ \text{coker}(\varphi) &= M'/\text{im}(\varphi) \\ \text{coim}(\varphi) &= M/\ker \varphi\end{aligned}$$

Definition 1.2.13. Let M be an R -module and I an ideal of R . Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I, m_i \in M \right\}$$

Proposition 1.2.14. Let M be an R -module. Then $R \rightarrow \text{End}(M)$ is a ring homomorphism.

1. $R \rightarrow \text{End}(M)$ factors through the natural surjection $R \rightarrow R/I$ if and only if $IM = 0$:

$$\begin{array}{ccc} R & \xrightarrow{\rho} & \text{End}(M) \\ & \searrow \pi_I & \nearrow \exists \rho_I \\ & R/I & \end{array}$$

Where ρ factors through π_I means $\exists \rho_I : R/I \rightarrow \text{End}(M)$ such that the diagram above commutes.

2. The module M/IM is naturally endowed with a structure of R/I -module.
3. Let $\varphi : M \rightarrow M'$ be an R -homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I -modules

$$M/IM \rightarrow M'/IM'.$$

4. If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.
5. $R^n/IR^n \cong (R/IR)^n$.

Proof. (1) Suppose first that ρ factors through π_I . Then there exists a ring homomorphism

$$\rho_I : R/I \rightarrow \text{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I : there exists a unique ring homomorphism

$$\rho_I : R/I \rightarrow \text{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M , we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M . Therefore M/IM is naturally an R/I -module.

(3) Let $\varphi : M \rightarrow M'$ be an R -module homomorphism. Take any element of IM , say

$$x = \sum_{j=1}^t a_j m_j \quad (a_j \in I, m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^t \varphi(a_j m_j) = \sum_{j=1}^t a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j \varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\bar{\varphi} : M/IM \rightarrow M'/IM', \quad \bar{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\bar{\varphi}$ is R/I -linear, since

$$\bar{\varphi}((r + I) \cdot (m + IM)) = \bar{\varphi}(rm + IM) = \varphi(rm) + IM' = r\varphi(m) + IM' = (r + I) \cdot \bar{\varphi}(m + IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \quad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \quad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i . Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \rightarrow M/IM$$

by

$$\Psi(m_1 + IM_1, \dots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I -linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i , and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. □

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Direct sum and direct product ($\bigoplus_{i \in I} M_i$ & $\prod_{i \in I} M_i$).*

Let $(M_i)_{i \in I}$ be a family of R -modules. The direct product of $(M_i)_{i \in I}$ is the module

$$\prod_{i \in I} M_i = \{\varphi : I \rightarrow \cup_{i \in I} M_i \mid \forall i \in I, \varphi(i) \in M_i\}$$

with the componentwise addition and scalar multiplication. i.e.

$$R \times \prod_{i \in I} M_i \rightarrow \prod_{i \in I} M_i, \quad (r, \varphi) \mapsto (r\varphi(i))_{i \in I}$$

The direct sum of $(M_i)_{i \in I}$ is the module

$$\bigoplus_{i \in I} M_i = \{\varphi \in \prod_{i \in I} M_i \mid \#\{i \in I \mid \varphi(i) \neq 0\} < \infty\}$$

Equal when $|I| < \infty$. In a module category, direct product = product, direct sum = coproduct.

Theorem 1.3.2. Suppose that $M_1, \dots, M_n$ are R -submodules of M in which $M = M_1 + \dots + M_n$, $I = [n]$. Then the followings are equivalent:

1. The map $\varphi : \sum_{i \in I} M_i \rightarrow M$ by $\varphi((m_i)_{i \in I}) = \sum_{i \in I} m_i, m_i \in M_i$ is an isomorphism.
2. Any element of M can be uniquely written as a sum of elements of M_i .
3. 0 can be uniquely written as a sum of elements of M_i . More precisely, $0 = 0 + \dots + 0$.
4. For $i \in I$, $M_i \cap (\sum_{j \neq i} M_j) = 0$.

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

(1) $\Rightarrow$ (2). If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \dots, m_n) \in \sum_{i=1}^n M_i$$

such that

$$m = \varphi(m_1, \dots, m_n) = m_1 + \dots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i .

(2) $\Rightarrow$ (3). Apply (2) to the element $0 \in M$.

(3) $\Rightarrow$ (4). Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j \neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j \neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \dots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) $\Rightarrow$ (1). Since

$$M = M_1 + \dots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \dots, m_n) = 0,$$

that is,

$$m_1 + \dots + m_n = 0 \quad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = - \sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i , so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module). *Let M be an R -module.*

1. $x_1, \dots, x_n \in M$ are linearly independent if

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = 0 \Rightarrow a_1 = a_2 = \dots = a_n = 0.$$

2. M is called free if it is generated by a linearly independent subset. This subset is called a basis of M .

3. If M is an R -module with a basis $x_1, \dots, x_n$, then $M = \langle x_1 \rangle \oplus \dots \oplus \langle x_n \rangle$.

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R -module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i \in I}$ be a basis of M . We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \dots, y_n$ of M . Each y_k is a finite R -linear combination of elements of B , so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \dots, y_n \in \sum_{i \in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^n Ry_k \subseteq \sum_{i \in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i \in I_0} Rx_i$. If $I \setminus I_0 \neq \emptyset$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i \in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \dots, x_r$ and $y_1, \dots, y_s$ be two bases of M . Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F -vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM , and $\overline{a}$ for the class of $a \in R$ in F . We claim that $\overline{x_1}, \dots, \overline{x_r}$ form an F -basis of M/JM .

Spanning. For any $m \in M$, write $m = \sum_{i=1}^r a_i x_i$. Then

$$\overline{m} = \sum_{i=1}^r \overline{a_i} \overline{x_i},$$

so $\overline{x_1}, \dots, \overline{x_r}$ span M/JM .

Linear independence. Suppose

$$\sum_{i=1}^r \overline{a_i} \overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^r a_i x_i \in JM$, so there exist $a'_1, \dots, a'_r \in J$ such that

$$\sum_{i=1}^r a_i x_i = \sum_{i=1}^r a'_i x_i.$$

Hence

$$\sum_{i=1}^r (a_i - a'_i) x_i = 0.$$

By linear independence of $x_1, \dots, x_r$, we get $a_i - a'_i = 0$ for all i , so $a_i \in J$, thus $\overline{a_i} = 0$ for

all i . Therefore $\overline{x_1}, \dots, \overline{x_r}$ are F -linearly independent, and form an F -basis of M/JM .

By the same argument, $\overline{y_1}, \dots, \overline{y_s}$ also form an F -basis of M/JM . Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. □

Chapter 2

Lecture 2

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. Let M be an R -module, $m \in M$, $Rm = \langle m \rangle = \{rm \mid r \in R\}$. Define the annihilator of m in R as:

$$\text{Ann}_R(m) = \{r \in R \mid rm = 0\}.$$

It is a left ideal of R .

Then we have $R/\text{Ann}_R(m) \cong Rm$ is an isomorphism. Which shows that all cyclic modules are isomorphic to a quotient of R .

Definition 2.1.2. An element $m \in M$ is called

- **torsion-free** if $\text{Ann}_R(m) = 0$.
- **torsion** if $\text{Ann}_R(m) \neq 0$.

A module is called torsion-free if all its elements except of zero are torsion-free.

Definition 2.1.3. The torsion part of M is defined as:

$$\text{Tor}(M) = \{m \in M \mid \text{Ann}_R(m) \neq 0\}.$$

Lemma 2.1.4. If R is an integral domain, then the torsion submodule of M is a submodule of M .

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\text{Ann}_R(0_M) = R \neq 0$, so $0_M \in \text{Tor}(M)$.

Closed under addition. Let $m, n \in \text{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that

$\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m+n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\text{Ann}_R(m+n) \neq 0$ and $m+n \in \text{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \text{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\text{Ann}_R(rm) \neq 0$ and $rm \in \text{Tor}(M)$.

Therefore $\text{Tor}(M)$ is a submodule of M . □

Example 2.1.5. (1) $0 \in M$ is torsion.

(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if ${}_R R$ is torsion-free. In this case $\text{Tor}({}_R R) = 0$, $\text{Tor}(R^n) = 0$ for all $n \geq 1$.

(3) If I is a nonzero ideal of R , then $\text{Tor}(R/I) = R/I$ where R/I is regarded as an R -module. If we regard R/I as an R/I -module, then it is torsion-free if and only if I is a prime ideal.

(4) More generally, let I be any non-zero ideal of R , and M be an R -module. Then $\text{Tor}(M/IM) = M/IM$ where M/IM is regarded as an R -module.

Remark 2.1.6. Let R be a commutative ring with 1. Let M be an R -module. Then $f : M \rightarrow M$ such that $f(m) = rm$ for all $m \in M$ and $r \in R$ is a homomorphism of R -modules. If moreover M is torsion-free, then f is a monomorphism.

Lemma 2.1.7. If R is an integral domain and M is an R -module, then

1. $\text{Tor}(\text{Tor}(M)) = \text{Tor}(M)$.
2. $\text{Tor}(M/\text{Tor}(M)) = 0$.

Proof.

1. Since $\text{Tor}(M)$ is a submodule of M , its torsion submodule is

$$\text{Tor}(\text{Tor}(M)) = \{ m \in \text{Tor}(M) \mid \text{Ann}_R(m) \neq 0 \}.$$

Every $m \in \text{Tor}(M)$ already satisfies $\text{Ann}_R(m) \neq 0$, so $\text{Tor}(\text{Tor}(M)) \subseteq \text{Tor}(M)$. Conversely, $\text{Tor}(M) \subseteq \text{Tor}(\text{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \text{Tor}(M)$, its annihilator computed in the submodule $\text{Tor}(M)$ is the same as $\text{Ann}_R(m) \neq 0$, so $m \in \text{Tor}(\text{Tor}(M))$. Hence $\text{Tor}(\text{Tor}(M)) = \text{Tor}(M)$.

2. Let $\pi : M \rightarrow M/\text{Tor}(M)$ be the canonical projection. Write $\bar{m} = \pi(m) = m +$

$\text{Tor}(M)$. Suppose $\bar{m} \in \text{Tor}(M/\text{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda \bar{m} = \bar{0} \quad \text{in } M/\text{Tor}(M).$$

This means $\lambda m \in \text{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \text{Tor}(M)$, so $\bar{m} = \bar{0}$. Therefore $\text{Tor}(M/\text{Tor}(M)) = 0$.

□

Definition 2.1.8. Let J be a subscript set, denote $R^J = \prod_{j \in J} R$. $R^{(J)} = \bigoplus_{j \in J} R$. Let M be an R -module. Let $x_j \in M$ where $j \in J$ be a family of elements of M . Then the submodule generated by $(x_j)_{j \in J}$ is denoted by $\sum_{j \in J} Rx_j$.

Corollary 2.1.9. We have a morphism of R -modules $\phi : R^{(J)} \rightarrow M$ such that $\phi(r_j)_{j \in J} = \sum_{j \in J} r_j x_j$ for all $(r_j)_{j \in J} \in R^{(J)}$.

- ϕ is injective if and only if $\{x_j\}_{j \in J}$ are linearly independent.
- ϕ is surjective if and only if $\{x_j\}_{j \in J}$ generate M .
- ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j \in J}$.

Lemma 2.1.10. Let L be a free R -module with basis $\{x_j\}_{j \in J}$. For any R -module M the following map:

$$\text{Hom}_R(L, M) \rightarrow M^J, \quad f \mapsto (f(x_j))_{j \in J}$$

is a bijection.

Proposition 2.1.11. Let L be a free R -module.

1. Let $\pi : X \rightarrow Y$ be an epimorphism of R -modules. For any R -module homomorphism $\varphi : L \rightarrow Y$ there exists a morphism $\tilde{\varphi} : L \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} & L & \\ \tilde{\varphi} \swarrow & \downarrow \varphi & \\ X & \xrightarrow{\pi} & Y \longrightarrow 0 \end{array}$$

2. In particular, for any R -module M any epimorphism $\pi : M \rightarrow L$ has a section, that is, there exists a morphism $\sigma : L \rightarrow M$ such that $\pi\sigma = \text{id}_L$. In that case σ is injective, $\text{im}(\sigma) = L$ and $M = \ker \pi \oplus \text{im}(\sigma)$.

$$0 \longrightarrow \ker \pi \longrightarrow M \xrightarrow[\sigma]{\varphi} L \longrightarrow 0$$

Proof. Let $\{e_j\}_{j \in J}$ be a basis of L .

(1) Since π is surjective, for each $j \in J$ there exists $\tilde{x}_j \in X$ such that $\pi(\tilde{x}_j) = \varphi(e_j)$. By the lemma above, the map $\text{Hom}_R(L, X) \rightarrow X^J$ given by $f \mapsto (f(e_j))_{j \in J}$ is a bijection. Hence there exists a unique R -module homomorphism $\tilde{\varphi} : L \rightarrow X$ such that $\tilde{\varphi}(e_j) = \tilde{x}_j$ for all $j \in J$. Then for each basis element e_j ,

$$\pi \circ \tilde{\varphi}(e_j) = \pi(\tilde{x}_j) = \varphi(e_j),$$

so $\pi \circ \tilde{\varphi}$ and φ agree on the basis $\{e_j\}_{j \in J}$. Since the basis generates L , we conclude $\pi \circ \tilde{\varphi} = \varphi$.

Remark 2.1.12. *The uniqueness appearing in the proof of the lifting property is not the uniqueness of the lift among all morphisms $\tilde{\varphi} : L \rightarrow X$ satisfying*

$$\pi \circ \tilde{\varphi} = \varphi.$$

Rather, it is only the uniqueness after a choice of lifts of the basis elements has been fixed.

Indeed, let L be a free R -module with basis $\{e_j\}_{j \in J}$. Since $\pi : X \rightarrow Y$ is surjective, for each $j \in J$ we may choose an element $\tilde{x}_j \in X$ such that

$$\pi(\tilde{x}_j) = \varphi(e_j).$$

By the universal property of a free module, or equivalently by the bijection

$$\text{Hom}_R(L, X) \xrightarrow{\sim} X^J, \quad f \mapsto (f(e_j))_{j \in J},$$

there exists a unique R -module homomorphism

$$\tilde{\varphi} : L \rightarrow X$$

such that

$$\tilde{\varphi}(e_j) = \tilde{x}_j \quad \text{for all } j \in J.$$

Thus the word “unique” here means: for the chosen family $(\tilde{x}_j)_{j \in J}$, there is a unique homomorphism realizing those values on the basis.

However, the elements $\tilde{x}_j$ themselves are generally not unique. If $\tilde{x}_j$ is one lift of $\varphi(e_j)$, then for any $k_j \in \ker \pi$, the element

$$\tilde{x}_j + k_j$$

is also a lift, since

$$\pi(\tilde{x}_j + k_j) = \pi(\tilde{x}_j) + \pi(k_j) = \varphi(e_j).$$

Different choices of lifts of the basis elements therefore give, in general, different morphisms $\tilde{\varphi} : L \rightarrow X$. Hence the proposition asserts only existence of a lift, not uniqueness.

Equivalently, if $\tilde{\varphi}_1, \tilde{\varphi}_2 : L \rightarrow X$ are two lifts of φ , then

$$\pi \circ \tilde{\varphi}_1 = \varphi = \pi \circ \tilde{\varphi}_2,$$

so

$$\pi \circ (\tilde{\varphi}_1 - \tilde{\varphi}_2) = 0.$$

Therefore

$$\text{im}(\tilde{\varphi}_1 - \tilde{\varphi}_2) \subseteq \ker \pi,$$

that is,

$$\tilde{\varphi}_1 - \tilde{\varphi}_2 \in \text{Hom}_R(L, \ker \pi).$$

Conversely, if $\delta \in \text{Hom}_R(L, \ker \pi)$, then

$$\pi \circ (\tilde{\varphi}_1 + \delta) = \pi \circ \tilde{\varphi}_1 = \varphi,$$

so $\tilde{\varphi}_1 + \delta$ is again a lift of φ . Thus the lift is unique if and only if

$$\text{Hom}_R(L, \ker \pi) = 0.$$

(2) Apply (1) with $X = M$, $Y = L$, and $\varphi = \text{id}_L$. Then there exists $\sigma : L \rightarrow M$ such that $\pi \circ \sigma = \text{id}_L$.

σ is injective: If $\sigma(l) = 0$, then $l = \text{id}_L(l) = \pi(\sigma(l)) = \pi(0) = 0$.

$M = \ker \pi \oplus \text{im}(\sigma)$: For any $m \in M$, write

$$m = (m - \sigma(\pi(m))) + \sigma(\pi(m)).$$

The second summand $\sigma(\pi(m)) \in \text{im}(\sigma)$. At first, $\pi(m - \sigma(\pi(m))) = \pi(m) - \pi\sigma(\pi(m)) = \pi(m) - \pi(m) = 0$, so it lies in $\ker \pi$. This shows $M = \ker \pi + \text{im}(\sigma)$.

For the directness, if $m \in \ker \pi \cap \text{im}(\sigma)$, write $m = \sigma(l)$ for some $l \in L$. Then $0 = \pi(m) = \pi(\sigma(l)) = l$, so $m = \sigma(0) = 0$. Hence the sum is direct. $\square$

Exercise 2.1.13. Let L be a free R -module. P a direct summand of L . Show that whenever $\pi : X \rightarrow Y$ is an epimorphism and $\varphi : P \rightarrow Y$ is a morphism, there exists a morphism $\tilde{\varphi} : P \rightarrow X$ such that $\varphi = \pi \circ \tilde{\varphi}$.

2.2 Tensor product

Let R be a commutative ring. M, N be R -modules. The tensor product $M \otimes_R N$ is a "linearize" of the bilinear map.

Definition 2.2.1. Let R be a commutative ring. M, N, K be R -modules. $f : M \times N \rightarrow K$ is a bilinear map if for all $r \in R$, $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, we have:

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$

- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
- $f(rm, n) = rf(m, n)$
- $f(m, rn) = rf(m, n)$

Example 2.2.2. If $f : M \times N \rightarrow K$ is a bilinear map, then $f(x, 0) = 0 = f(0, y)$ for all $x \in M, y \in N$. $f(-x, y) = -f(x, y)$ and $f(x, -y) = -f(x, y)$ for all $x \in M, y \in N$.

Definition 2.2.3. The tensor product of M and N is a pair $(M \otimes_R N, t_{M,N})$ where $M \otimes_R N$ is an R -module and $t_{M,N} : M \times N \rightarrow M \otimes_R N$ is a bilinear map usually denoted by $t_{M,N}(m, n) = m \otimes n$ which satisfies the following universal properties: Whenever X is an R -module and $f : M \times N \rightarrow X$ is a bilinear map, there exists a unique R -module homomorphism $\tilde{f} : M \otimes_R N \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & X \\ t_{M,N} \downarrow & \nearrow \tilde{f} & \\ M \otimes_R N & & \end{array}$$

Theorem 2.2.4. The tensor product of M and N exists and is unique up to isomorphism.

Proof. We prove the uniqueness at first.

The tensor product is unique up to isomorphism. Suppose (T_1, t_1) and (T_2, t_2) are two tensor products of M and N . By the universal property, there exists a unique R -module homomorphism $\varphi : T_1 \rightarrow T_2$ such that $t_2 = \varphi \circ t_1$ and a unique R -module homomorphism $\psi : T_2 \rightarrow T_1$ such that $t_1 = \psi \circ t_2$. Then we have:

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2$$

$$\psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

Since such $\varphi \circ \psi$ and $\psi \circ \varphi$ are unique. It forces $\varphi \circ \psi = \text{id}_{T_2}$ and $\psi \circ \varphi = \text{id}_{T_1}$. Hence φ and ψ are isomorphisms. Therefore $T_1 \cong T_2$.

Now we prove the existence.

Let F be a free R -module with basis $M \times N$. i.e. $F = \text{span}_R(e_{m_i, n_i})_{m_i \in M, n_i \in N}$. Let B be the submodule of F generated by the elements of the following forms:

- $e_{m_1+m_2, n} - e_{m_1, n} - e_{m_2, n}$
- $e_{m, n_1+n_2} - e_{m, n_1} - e_{m, n_2}$
- $e_{rm, n} - re_{m, n}$

- $e_{m,rn} - re_{m,n}$

Where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, $r \in R$. Define $M \otimes_R N = F/B$, $t_{M,N} : M \times N \rightarrow M \otimes_R N$ by $t_{M,N}(m, n) = e_{m,n} + B$. It is easy to check that $t_{M,N}$ is a well-defined bilinear map.

We only need to show that $(M \otimes_R N, t_{M,N})$ satisfies the universal property of tensor product:

Let X be an R -module and $f : M \times N \rightarrow X$ a bilinear map. Since F is a free R -module with basis $\{e_{m,n}\}_{(m,n) \in M \times N}$, by the universal property of the free module, there exists a unique R -module homomorphism $\hat{f} : F \rightarrow X$ such that $\hat{f}(e_{m,n}) = f(m, n)$ for all $(m, n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear:

$$\begin{aligned}\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1 + m_2, n) - f(m_1, n) - f(m_2, n) = 0, \\ \hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m, n_1 + n_2) - f(m, n_1) - f(m, n_2) = 0, \\ \hat{f}(e_{rm,n} - re_{m,n}) &= f(rm, n) - rf(m, n) = 0, \\ \hat{f}(e_{m,rn} - re_{m,n}) &= f(m, rn) - rf(m, n) = 0.\end{aligned}$$

Since B is generated by elements of the above forms and $\ker \hat{f}$ is a submodule, we have $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R -module homomorphism $\tilde{f} : M \otimes_R N = F/B \rightarrow X$ such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi : F \rightarrow F/B$ is the canonical projection. In particular, $\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m, n)$, so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g : M \otimes_R N \rightarrow X$ is another R -module homomorphism with $g \circ t_{M,N} = f$. Then $g(m \otimes n) = f(m, n) = \tilde{f}(m \otimes n)$ for all $m \in M, n \in N$. Since elements of the form $m \otimes n$ generate $M \otimes_R N$ as an R -module, $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N . □

Definition 2.2.5. The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R -module.

Example 2.2.6. Fix $m \in M$. The map $N \rightarrow M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R -modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N / \text{Ann}_R(m)N$.

Proposition 2.2.7. We have the following isomorphisms:

1. $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ where $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.
2. $M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ where $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.
3. $M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ where $m_1 \otimes (m_2 \oplus m_3) \mapsto$

$(m_1 \otimes m_2) + (m_1 \otimes m_3)$. And more generally,

$$M_1 \otimes_R \left(\bigoplus_{i=1}^n M_i \right) \cong \bigoplus_{i=1}^n (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ where $r \otimes m \mapsto rm$. And more generally, $R^{(I)} \otimes_R M \cong M^{(I)}$ where $(r_i)_{i \in I} \otimes m \mapsto (r_i m)_{i \in I}$.

Proof. We prove (1): $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3 , define $f_{m_3} : M_1 \times M_2 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2) , so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R -module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{array}{ccc} M_1 \times M_2 & \xrightarrow{f_{m_3}} & M_1 \otimes_R (M_2 \otimes_R M_3) \\ \downarrow t_{M_1, M_2} & \nearrow \tilde{f}_{m_3} & \\ M_1 \otimes_R M_2 & & \end{array}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R -module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned} g(m_1 \otimes m_2, m_3 + m'_3) &= m_1 \otimes (m_2 \otimes (m_3 + m'_3)) \\ &= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m'_3) \\ &= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m'_3) \\ &= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m'_3), \\ g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\ &= m_1 \otimes (r(m_2 \otimes m_3)) \\ &= r(m_1 \otimes (m_2 \otimes m_3)) \\ &= r g(m_1 \otimes m_2, m_3). \end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R -module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{array}{ccc} (M_1 \otimes_R M_2) \times M_3 & \xrightarrow{g} & M_1 \otimes_R (M_2 \otimes_R M_3) \\ \downarrow t & \nearrow \varphi & \\ (M_1 \otimes_R M_2) \otimes_R M_3 & & \end{array}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R -module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \rightarrow (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules, $\psi \circ \varphi = \text{id}$ and $\varphi \circ \psi = \text{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{array}{ccc} & M_1 \times M_2 \times M_3 & \\ \swarrow & & \searrow \\ (M_1 \otimes_R M_2) \otimes_R M_3 & \xrightleftharpoons[\psi]{\varphi} & M_1 \otimes_R (M_2 \otimes_R M_3) \end{array}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. $\square$

Theorem 2.2.8. Let M, N, P be R -modules. We have a canonical isomorphism:

$$\phi_{M,N,P} : \text{Hom}_R(M \otimes_R N, P) \rightarrow \text{Hom}_R(M, \text{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R -module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$ -module isomorphism.

Proof. Let $f \in \text{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \rightarrow P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \text{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned} \phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\ &= f(m \otimes n_1) + f(m \otimes n_2), \\ \phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\ &= rf(m \otimes n) = r\phi_{M,N,P}(f)(m)(n). \end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \text{Hom}_R(M, \text{Hom}_R(N, P))$: for $m_1, m_2 \in M$, $r \in R$, and

any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= rf(m \otimes n) \\
&= (r \phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r \phi_{M,N,P}(f)(m)$, confirming R -linearity.

Let $g \in \text{Hom}_R(M, \text{Hom}_R(N, P))$. Define $\hat{g} : M \times N \rightarrow P$ by $\hat{g}(m, n) = g(m)(n)$. We check that $\hat{g}$ is R -bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n), \\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2), \\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r \hat{g}(m, n), \\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r \hat{g}(m, n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R -module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \rightarrow P$ such that $\psi_{M,N,P}(g)(m \otimes n) = \hat{g}(m, n)$:

$$\begin{array}{ccc}
M \times N & \xrightarrow{\hat{g}} & P \\
t_{M,N} \downarrow & \nearrow \psi_{M,N,P}(g) & \\
M \otimes_R N & &
\end{array}$$

For any $f \in \text{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \text{Hom}_R(M, \text{Hom}_R(N, P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m , hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\text{Hom}_R(M \otimes_R N, P)$ and $\text{Hom}_R(M, \text{Hom}_R(N, P))$ are abelian groups under point-

wise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \text{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}\phi_{M,N,P}(f_1 + f_2)(m)(n) &= (f_1 + f_2)(m \otimes n) \\ &= f_1(m \otimes n) + f_2(m \otimes n) \\ &= \phi_{M,N,P}(f_1)(m)(n) + \phi_{M,N,P}(f_2)(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(f_1 + f_2) = \phi_{M,N,P}(f_1) + \phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$ -module isomorphism.

If R is commutative, $\text{Hom}_R(M \otimes_R N, P)$ carries an R -module structure via $(r \cdot f)(x) = rf(x)$, and similarly for $\text{Hom}_R(M, \text{Hom}_R(N, P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m \otimes n) \\ &= rf(m \otimes n) \\ &= r \phi_{M,N,P}(f)(m)(n) \\ &= (r \phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r \phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R -module isomorphism.

If R is not commutative, $\text{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R -module structure (scalar multiplication by r need not commute with R -linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$ -module isomorphism. $\square$

Definition 2.2.9. Let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be R -module homomorphisms. The tensor product of f and g is the R -module homomorphism:

$$f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$$

defined by $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Definition 2.2.10. Let $M_i, i \in I$ be a family of R -modules. Use the following notation:

$$L^n(M_1, \dots, M_n; N) = \{f : M_1 \times \dots \times M_n \rightarrow N \mid f \text{ is } n\text{-linear}\}$$

In particular,

$$L^2(M_1, M_2; N) \cong L(M_1 \otimes M_2, N) \cong \text{Hom}_R(M_1 \otimes M_2, N)$$

or equivalently,

$$L^2(M_1 \otimes M_2, N) \cong L(M_1, L(M_2, N))$$

Proposition 2.2.11. Let M, N be two finitely generated free R -modules. Denote

$$M^\vee = \text{Hom}_R(M, R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \text{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R -module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \rightarrow \text{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \rightarrow \text{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m) n, \quad f \in M^\vee, n \in N, m \in M.$$

We check R -bilinearity. For $f_1, f_2 \in M^\vee$, $n, n_1, n_2 \in N$, $m \in M$, and $r \in R$:

$$\begin{aligned} \beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m) n = f_1(m) n + f_2(m) n \\ &= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\ \beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m) n_1 + f(m) n_2 \\ &= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\ \beta(rf, n)(m) &= (rf)(m) n = r f(m) n = r \beta(f, n)(m), \\ \beta(f, rn)(m) &= f(m) (rn) = r f(m) n = r \beta(f, n)(m). \end{aligned}$$

By the universal property of the tensor product, there exists a unique R -module homomorphism

$$\Phi : M^\vee \otimes_R N \rightarrow \text{Hom}_R(M, N), \quad \Phi(f \otimes n)(m) = f(m) n.$$

Now let $\{e_1, \dots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \dots, e_m^*\}$ be the dual basis, i.e. $e_i^* \in M^\vee$ with $e_i^*(e_j) = \delta_{ij}$. Define

$$\Psi : \text{Hom}_R(M, N) \rightarrow M^\vee \otimes_R N, \quad \Psi(g) = \sum_{i=1}^m e_i^* \otimes g(e_i).$$

We verify Ψ is an R -module homomorphism: for $g_1, g_2 \in \text{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned} \Psi(g_1 + g_2) &= \sum_{i=1}^m e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^m e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\ &= \sum_{i=1}^m e_i^* \otimes g_1(e_i) + \sum_{i=1}^m e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\ \Psi(rg) &= \sum_{i=1}^m e_i^* \otimes (rg)(e_i) = \sum_{i=1}^m e_i^* \otimes r g(e_i) \\ &= r \sum_{i=1}^m e_i^* \otimes g(e_i) = r \Psi(g). \end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \text{Hom}_R(M, N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}\Phi(\Psi(g))(m) &= \Phi\left(\sum_{i=1}^m e_i^* \otimes g(e_i)\right)\left(\sum_j r_j e_j\right) \\ &= \sum_{i=1}^m e_i^*\left(\sum_j r_j e_j\right)g(e_i) = \sum_{i=1}^m r_i g(e_i) \\ &= g\left(\sum_{i=1}^m r_i e_i\right) = g(m).\end{aligned}$$

So $\Phi \circ \Psi = \text{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}\Psi(\Phi(f \otimes n)) &= \sum_{i=1}^m e_i^* \otimes \Phi(f \otimes n)(e_i) = \sum_{i=1}^m e_i^* \otimes f(e_i) n \\ &= \sum_{i=1}^m f(e_i) (e_i^* \otimes n) = \left(\sum_{i=1}^m f(e_i) e_i^*\right) \otimes n = f \otimes n,\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^m f(e_i) e_i^*$ (expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi \circ \Phi$ is R -linear, we have $\Psi \circ \Phi = \text{id}$.

Therefore Φ is an R -module isomorphism, giving $M^\vee \otimes_R N \cong \text{Hom}_R(M, N)$. $\square$

Proposition 2.2.12. *Assume that R is a subring of S , and M is an R -module. Then the bilinear map:*

$$S \times (S \otimes_R M) \rightarrow S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S -module on the R -module $S \otimes_R M$.

More generally, let $f : R \rightarrow T$ be a ring homomorphism. And view T as an R -module via $R \times T \rightarrow T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \rightarrow T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T -module on the R -module $T \otimes_R M$.

Proposition 2.2.13. *Let I be an ideal of R . Let M be an R -module. Then the morphism of R -modules:*

$$M \rightarrow (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R -modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I -modules.

2.3 Balanced product

Definition 2.3.1. Let R be a ring (not necessarily commutative). Let M_R be a right R -module and ${}_R N$ be a left R -module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \rightarrow P$ is a bilinear map satisfies:

- $f(rm, n) = f(m, rn)$ for all $r \in R, m \in M, n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ for all $m, m' \in M, n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ for all $m \in M, n, n' \in N$.

Definition 2.3.2. A tensor product of M_R and ${}_R N$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and ${}_R N$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \rightarrow P$ such that the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{t_{M,N}} & M \otimes_R N \\ f \downarrow & \tilde{f} \swarrow & \\ P & & \end{array}$$

Chapter 3

Homework

3.1 Homework Week 1

Exercise 3.1.1. Let R be a ring, and let M be an R -module. Suppose that S is a ring and

$$\theta : S \rightarrow R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S -module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_Sx = \theta(1_S)x = 1_Rx = x$

Hence M is an S -module. □

Exercise 3.1.2. Let R be a ring, and let M and N be two right R -modules. Let $\text{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\text{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$, define

$$(\varphi a)(x) = \varphi(xa), \quad \forall x \in M.$$

Prove that $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module.

Proof. This action is well-defined. $\forall r \in R$, $\varphi r \in \text{Hom}_{\mathbb{Z}}(M, N)$ for all $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$ as $(\varphi r)(x) = \varphi(xr) \in N$ for all $x \in M$. We need to verify the module axioms, for all $r_1, r_2 \in R$ and $\varphi, \psi \in \text{Hom}_{\mathbb{Z}}(M, N)$, $\forall x \in M$:

- $((\varphi + \psi)r)(x) = (\varphi + \psi)(xr) = \varphi(xr) + \psi(xr) = (\varphi r)(x) + (\psi r)(x)$
- $(\varphi(r_1 + r_2))(x) = \varphi(x(r_1 + r_2)) = \varphi(xr_1 + xr_2) = \varphi(xr_1) + \varphi(xr_2) = (\varphi r_1)(x) + (\varphi r_2)(x)$
- $((\varphi r_1)r_2)(x) = (\varphi r_2)(xr_1) = \varphi(xr_2r_1) = (\varphi r_2r_1)(x)$
- $(\varphi 1_R)(x) = \varphi(x 1_R) = \varphi(x)$

Hence $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module. □

Exercise 3.1.3. Let R be a ring, and let M be an R -module. Denote

$$\text{Ann}_R M = \{r \in R \mid rm = 0, \forall m \in M\}.$$

We call M a faithful R -module if $\text{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\text{Ann}_R M$ on M by

$$\bar{r}x = rx,$$

where $x \in M$, $r \in R$, and $\bar{r} = r + \text{Ann}_R M$. Prove that M is a faithful $R/\text{Ann}_R M$ -module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\text{Ann}_R M.$$

Then $r - r' \in \text{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R -module structure on M .

Now we prove faithfulness. Suppose $\bar{r} \in R/\text{Ann}_R M$ acts trivially on M , that is,

$$\bar{r}x = 0 \quad \forall x \in M.$$

By definition, this means

$$rx = 0 \quad \forall x \in M.$$

Therefore $r \in \text{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\text{Ann}_{R/\text{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\text{Ann}_R M$ -module. □

Exercise 3.1.4. Let $\varphi : M \rightarrow M$ be a homomorphism of R -modules such that

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker \varphi \oplus \text{im } \varphi.$$

Proof. Let $x \in M$. Then $x = \varphi(x) + (x - \varphi(x))$. Where $\varphi(x) \in \text{im } \varphi$ and $x - \varphi(x) \in \ker \varphi$. Hence $M = \text{im } \varphi + \ker \varphi$. We only need to show that $\ker \varphi \cap \text{im } \varphi = \{0\}$. Suppose $x \in \ker \varphi \cap \text{im } \varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Then $\varphi(x) = \varphi\varphi(y) = \varphi(y) = x = 0$, so $x = 0$. Hence $\ker \varphi \cap \text{im } \varphi = \{0\}$. $\square$

Exercise 3.1.5. Show that any R -module is a homomorphism image of some free R -module.

Proof. Let M be a left R -module. Consider the free R -module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M . Define an R -module homomorphism

$$\pi : F \rightarrow M$$

by

$$\pi(e_x) = x \quad (x \in M),$$

and extend R -linearly. Thus

$$\pi\left(\sum_{i=1}^n r_i e_{x_i}\right) = \sum_{i=1}^n r_i x_i.$$

This is well-defined and R -linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F . Equivalently,

$$M \cong F / \ker \pi.$$

$\square$

Exercise 3.1.6. Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$ -module?

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$ -module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n \frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$ -module. $\square$

3.2 Homework Week 2

Exercise 3.2.1. Let R be a commutative ring, and suppose that φ is an endomorphism of the free R -module R^m . Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?

Proof. Let $\{e_1, \dots, e_m\}$ be the standard basis of R^m , and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij}e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R . It follows that A is invertible (with $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective (e.g. $1 \notin \operatorname{im} \varphi$). $\square$

Exercise 3.2.2. Let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that:

1. φ is a monomorphism if and only if for any R -module T and any two R -module homomorphisms $\xi, \eta : T \rightarrow M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;
2. φ is an epimorphism if and only if for any R -module S and any two R -module homomorphisms $\rho, \theta : N \rightarrow S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.

Proof.

1. $(\Rightarrow)$ Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \rightarrow M$. Then for every $t \in T$, $\varphi(\xi(t)) = \varphi(\eta(t))$, so $\xi(t) - \eta(t) \in \ker \varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all t , i.e. $\xi = \eta$.
 $(\Leftarrow)$ We show $\ker \varphi = \{0\}$. Let $T = \ker \varphi$, $\xi : T \hookrightarrow M$ be the inclusion, and $\eta = 0 : T \rightarrow M$ the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker \varphi = \{0\}$, so φ is injective.
2. $(\Rightarrow)$ Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then $\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n)$. Hence $\rho = \theta$.
 $(\Leftarrow)$ We show φ is surjective. Let $S = N/\operatorname{im} \varphi$, $\rho : N \rightarrow S$ the canonical projection, and $\theta = 0 : N \rightarrow S$. For any $m \in M$, $\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m))$, so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \rightarrow N/\operatorname{im} \varphi$ is the zero map. This means $N = \operatorname{im} \varphi$, so φ is surjective.

$\square$

Exercise 3.2.3. Let M and M_i ($1 \leq i \leq n$) be R -modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{im} \varphi_i \simeq M_i$, $\varphi_i\varphi_j = 0$ for $i \neq j$, $\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \dots + \varphi_n = \operatorname{id}_M$.

Proof. ($\Rightarrow$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \text{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1} \iota_i \pi_i f f^{-1} \iota_i \pi_i f = f^{-1} \iota_i (\pi_i \iota_i) \pi_i f = f^{-1} \iota_i \pi_i f = \varphi_i$, since $\pi_i \iota_i = \text{id}_{M_i}$.
- For $i \neq j$: $\varphi_i \varphi_j = f^{-1} \iota_i (\pi_i \iota_j) \pi_j f = 0$, since $\pi_i \iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1} (\sum_{i=1}^n \iota_i \pi_i) f = f^{-1} \circ \text{id} \circ f = \text{id}_M$, since $\sum_i \iota_i \pi_i = \text{id}_{\bigoplus M_j}$.
- $\text{im } \varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \rightarrow \bigoplus_{i=1}^n \text{im } \varphi_i$ by $g(x) = (\varphi_1(x), \dots, \varphi_n(x))$.

Surjective: for any $(y_1, \dots, y_n) \in \bigoplus \text{im } \varphi_i$, let $x = y_1 + \dots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^n \varphi_j(y_i) = \sum_{i=1}^n \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j \varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \dots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i , so $x = \text{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\text{im } \varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 3.2.4.

1. Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.
2. Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.
3. Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow 0.$$

Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m, n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m, n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m, n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$ with A :

$$A \xrightarrow{\cdot n} A \rightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow 0.$$

By right-exactness, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^s \mathbb{Z}/d_i\mathbb{Z}, \quad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^u \mathbb{Z}/e_j\mathbb{Z}.$$

Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^u (\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^s (\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^s \bigoplus_{j=1}^u \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

□

Exercise 3.2.5. Let $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that

$$\text{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\text{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\text{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. □

Exercise 3.2.6. Let V_1 and V_2 be two vector spaces over a field F with dimension n and m respectively. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2 respectively. Suppose that $\mathcal{A}$ (resp. $\mathcal{B}$) is an F -linear map of V_1 (resp. V_2), and it is represented by the matrix $A = (a_{ij})_{n \times n}$ (resp. $B = (b_{ij})_{m \times m}$) with respect to the above bases. Prove that

$$\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$$

is an F -basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F -linear map $\mathcal{A} \otimes \mathcal{B}$ under this basis.

Proof. Step 1: $\{\epsilon_i \otimes \eta_j\}$ is a basis.

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i \epsilon_i$ and $w = \sum_j b_j \eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q) , let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals: $f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v)g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F -linear map $\Phi_{pq} : V_1 \otimes_F V_2 \rightarrow F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \leq i \leq n, 1 \leq j \leq m\}$ is an F -basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathcal{A} \otimes \mathcal{B}$.

Order the basis lexicographically: $\epsilon_1 \otimes \eta_1, \epsilon_1 \otimes \eta_2, \dots, \epsilon_1 \otimes \eta_m, \epsilon_2 \otimes \eta_1, \dots, \epsilon_n \otimes \eta_m$. Since $\mathcal{A}(\epsilon_i) = \sum_{k=1}^n a_{ki}\epsilon_k$ and $\mathcal{B}(\eta_j) = \sum_{l=1}^m b_{lj}\eta_l$, we have

$$(\mathcal{A} \otimes \mathcal{B})(\epsilon_i \otimes \eta_j) = \mathcal{A}(\epsilon_i) \otimes \mathcal{B}(\eta_j) = \sum_{k=1}^n \sum_{l=1}^m a_{ki}b_{lj}(\epsilon_k \otimes \eta_l).$$

The entry in row (k, l) and column (i, j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□